

6CS012 – Artificial Intelligence and Machine Learning

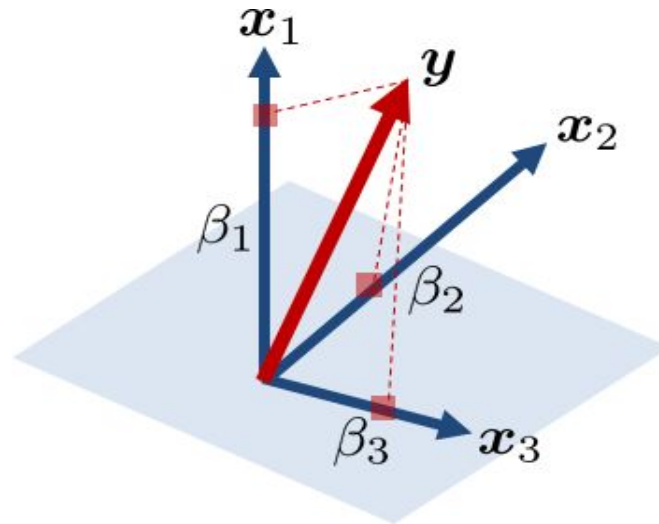


Image from Stanley Chan Book: Introduction to Probability for Data Science.

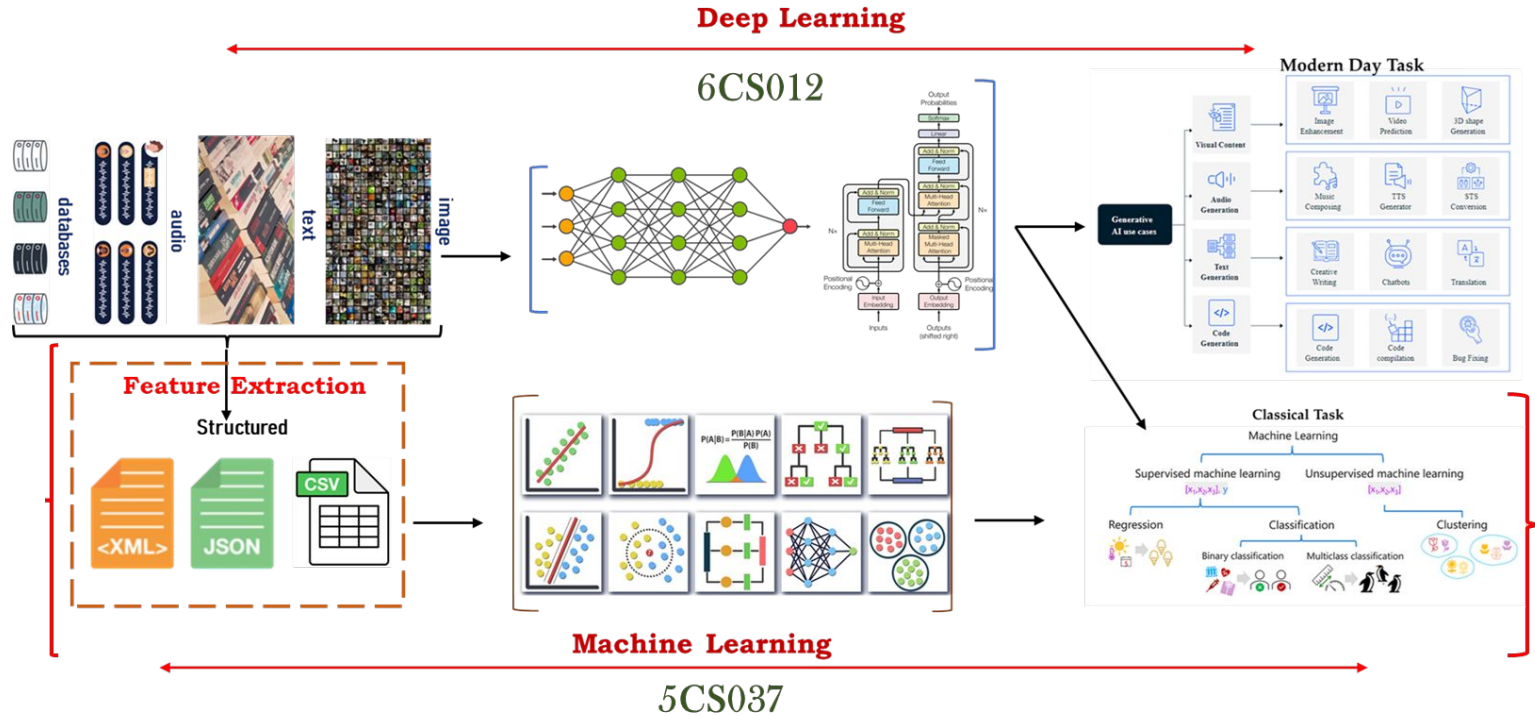
Foundational Math Skills for AI and ML

A quick revision on Linear Algebra

About the module!

What the module is about?

Difference in Approach



Learning Outcomes!!!

- Review and revise some fundamental concepts from Mathematics – **Linear Algebra and Derivative** we will be using through out the course.
 - **Cautions !!!**
 - We will omit many important topics in Linear algebra and Matrix Calculus, which we believe are not essential for understanding deep learning.
- In Particular we will discuss:
 - **Why do we need Linear Algebra for Machine/Deep Learning?**
 - **{Almost}** Everything we need to know about vector and matrices for Machine/Deep Learning.

Introduction to Linear Algebra

Why to study **Linear Algebra** for **Machine Learning**?

What is Linear Algebra?

- **Linear algebra** is a branch of mathematics focused on **vectors**, **vector spaces**, and **linear equations**.
- It helps us understand how to work with multiple variables in a structured way.
- **Key Concepts**

- **Linear Equations:** Relationships between variables, e.g.,

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

- **Vectors and Matrices:** Used to represent data and transformations.
- **Transformations:** Operations that move or scale data within vector spaces.

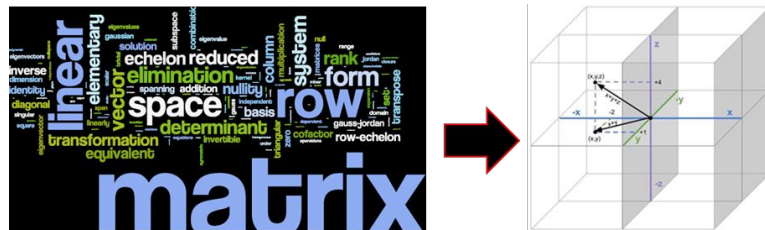


Fig: What is Linear Algebra?

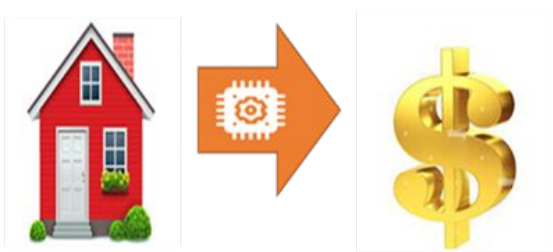
Image: somewhere from web compiled by siman

Why Linear Algebra for Machine Learning?

- **Representation of Data:**

- In machine learning, data is typically **represented** as **vectors** and **matrices**.
- For example, a dataset might be **stored as a matrix** where each row is a data point (vector), and each column is a feature.

Task: House Price Prediction.



Area	Rooms	Price
1080	8	1,00,000.00
1200	10	1,50,000.00

How would you represent this, for computer?
Matrix.

$$\begin{bmatrix} 1080 & 8 \\ 1200 & 10 \end{bmatrix} \begin{bmatrix} 1,00,000 \\ 1,50,000 \end{bmatrix}$$

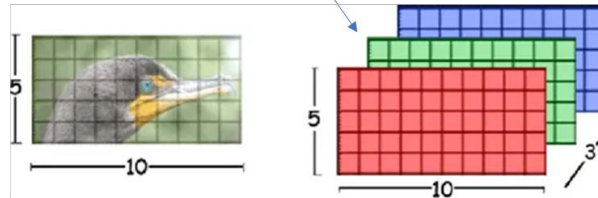
Why Linear Algebra for Machine Learning?

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- For example, a dataset might be represented as a matrix where each row is a data point (vector), and each column is a feature.



A gray scale image is represented with single matrix {R,G,B}



A color image is represented with three matrix {R,G,B}

Why Linear Algebra for Machine Learning?

- **Efficient Computing:**

- Matrix operations allow for efficient computations on large datasets.
- Libraries like **NumPy**, **TensorFlow**, and **PyTorch** leverage **linear algebra** for operations on large matrices and tensors **{Vectorizations}**, which makes **machine learning models faster** and more **scalable**.

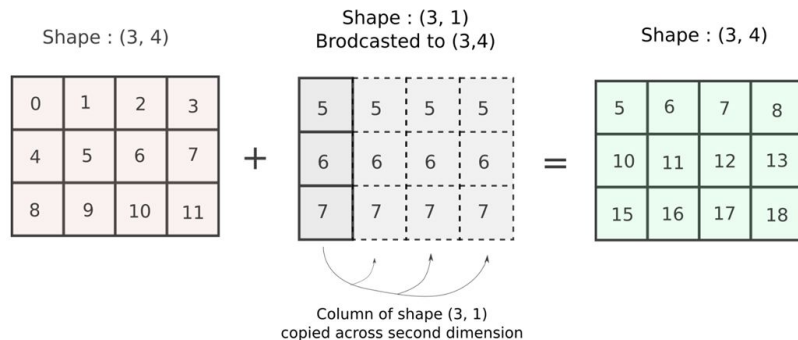


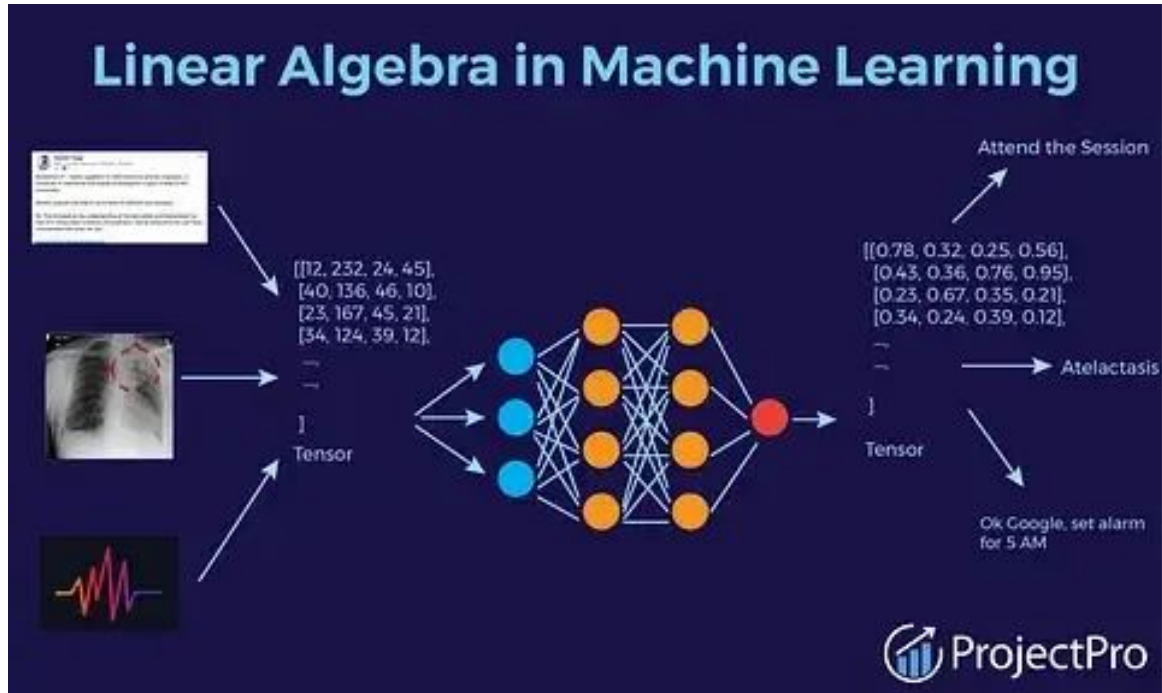
Fig: Idea of Vectorizations.

Why Linear Algebra for Machine Learning?

- **Understanding {Machine Learning} Algorithms:**

- **Training** machine learning models often involves **solving systems of linear equations**.
 - **Linear algebra** provides the **necessary tools** to solve these systems efficiently.
- Many machine learning algorithms are based on linear algebra concepts.
 - **Linear Regression** : Finding the best fitting hyperplane
 - **Support Vector Machines (SVM)**: Dot Product to measure Similarity
 - **Neural Network**: Matrix Multiplication for forward and backward propagation

Summary : Linear Algebra for Machine Learning.



Review: What are Vectors?

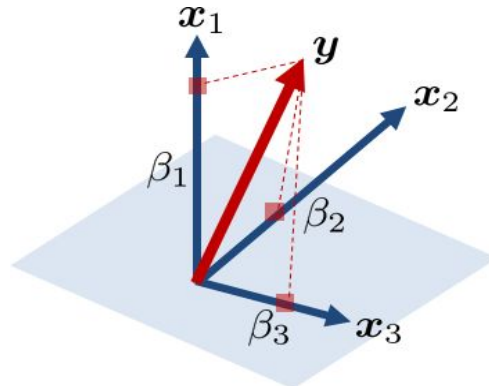
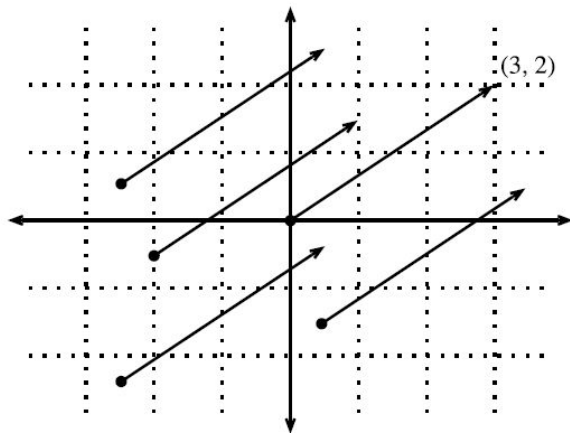


Image from Stanley Chan Book: Introduction to Probability for Data Science.

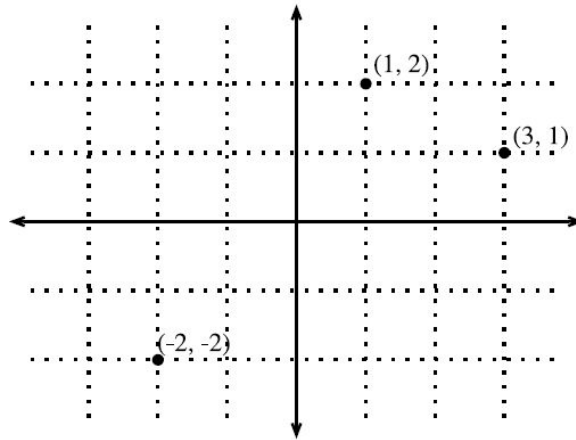
Direction in space

- E.g., the vector $\vec{v} = [3, 2]^T$ has a direction of 3 steps to the right and 2 steps up
- The notation $\vec{v}$ is used to indicate that the vectors have a direction
- All vectors in the figure have the same direction



Point in space

- E.g., in 2D we can visualize the data points with respect to a coordinate



In Machine Learning

The geometric interpretation of vectors as points in space allow us to consider a training set of input examples in ML as a **collection of points in space**

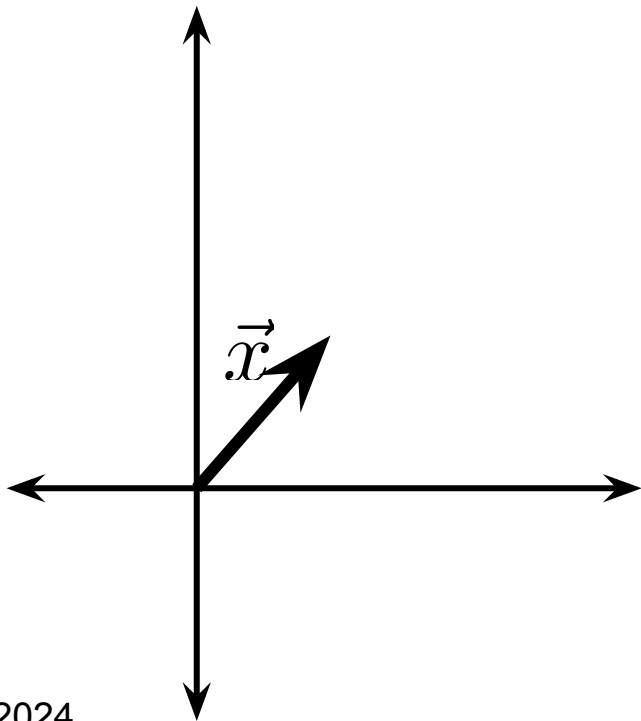
So, for eg.

classification can be viewed as discovering how to separate two clusters of points/vectors belonging to different classes

- Rather than distinguishing images containing cars, planes, buildings etc

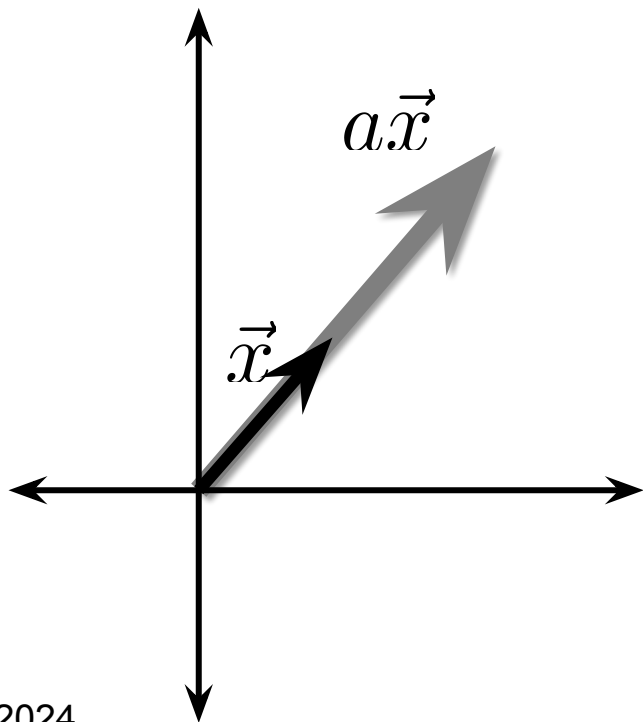


Scalars?

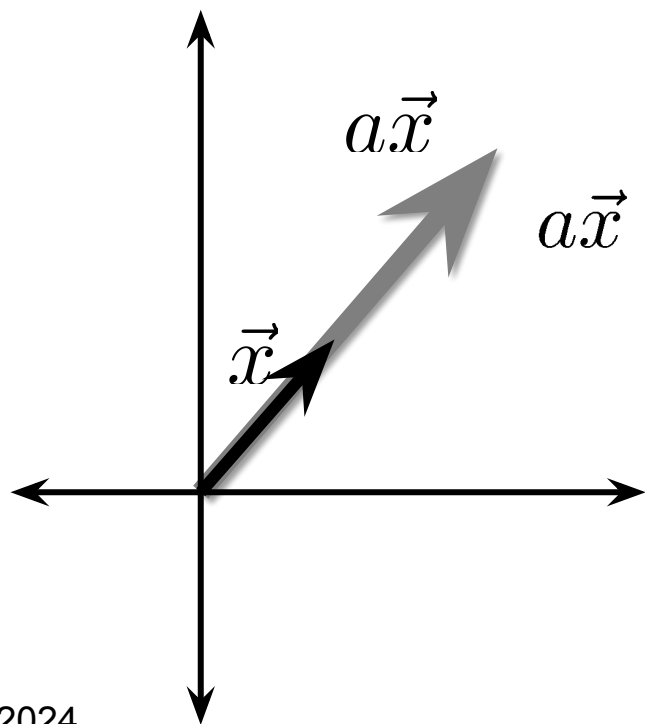


- A number real or Complex.
- Scales the vector by the number.

Scalar times vector



Scalar times vector


$$a\vec{x} = a \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} = \begin{pmatrix} ax_1 \\ ax_2 \\ \vdots \\ ax_N \end{pmatrix}$$

Operations on Vectors

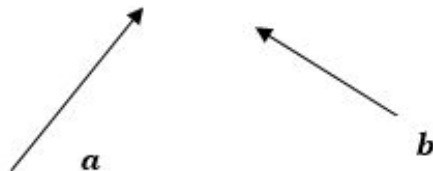
Addition on Vectors

Add element-wise.

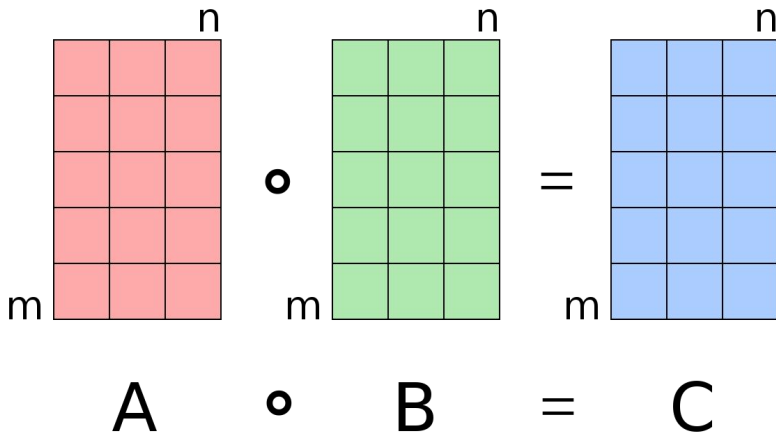
Can only add vectors of equal dimensions.

Associative and commutative

$$\begin{bmatrix} 3 \\ -6 \\ 7 \end{bmatrix} + \begin{bmatrix} -1 \\ 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$$



Element-by-element product (Hadamard product)

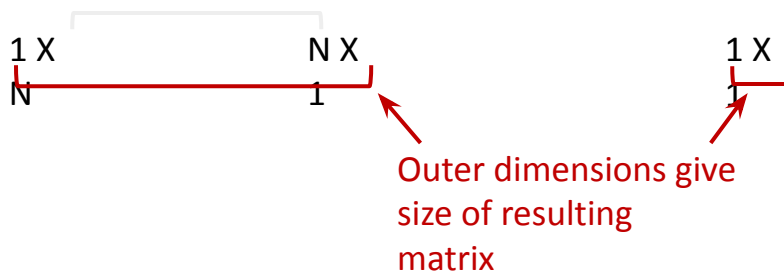
$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \cdot * \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ a_2 b_2 \end{pmatrix}$$


$A \circ B = C$

Dot product (inner product)

- Scalar Product of vectors $\vec{x} \cdot \vec{y} =$

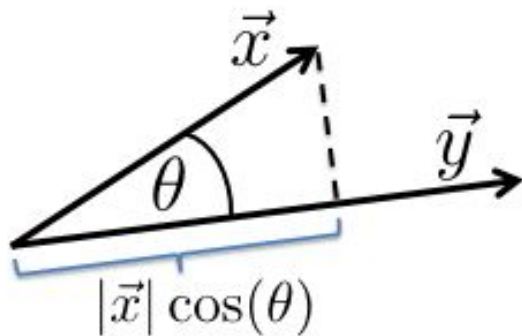
$$(x_1 \quad x_2 \quad \cdots \quad x_N) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = x_1 y_1 + x_2 y_2 + \cdots + x_N y_N$$



Geometric Intuition

Takes 2 vectors of the same dimension, returns a scalar

A measure of the “alignment” between two vectors, scaled by the lengths.



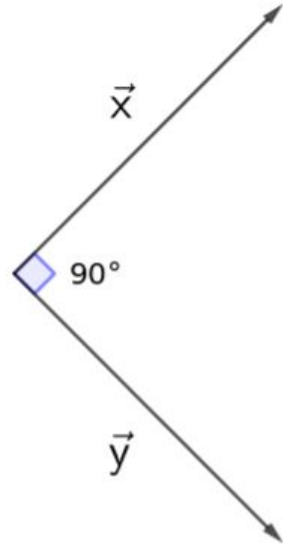
$$\vec{x} \cdot \vec{y} = |\vec{x}| |\vec{y}| \cos(\theta)$$

Geometric Intuition

Question

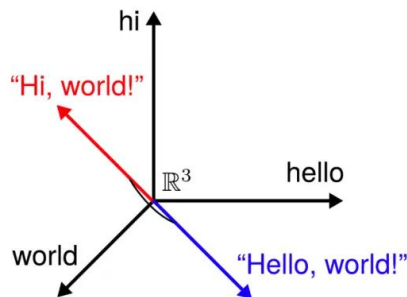
What happens when two vectors are orthogonal?

Dot product?

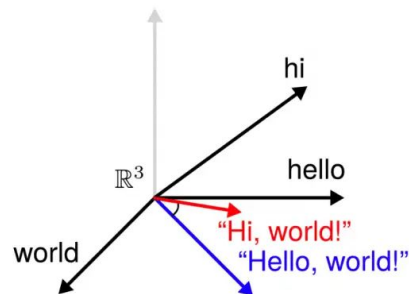


Geometric Intuition

Also, in ML the term $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ is sometimes employed as a measure of closeness of two vectors/data instances, and it is referred to as **cosine similarity**



Cosine Similarity



Soft Cosine Measure

Source: https://github.com/RaRe-Technologies/gensim/blob/develop/docs/notebooks/soft_cosine_tutorial.ipynb

Properties of a vector

- Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors of the same dimension and let α and β be scalars; then the following properties hold:

1. Commutativity

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$

$$\alpha \mathbf{x} = \mathbf{x} \alpha$$

2. Associativity

$$(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{y} + (\mathbf{x} + \mathbf{z})$$

$$\alpha(\beta \mathbf{x}) = (\alpha\beta)\mathbf{x}$$

3. Distributivity

$$\alpha(\mathbf{x} + \mathbf{y}) = \alpha \mathbf{x} + \alpha \mathbf{y}$$

$$(\alpha + \beta)\mathbf{x} = \alpha \mathbf{x} + \beta \mathbf{x}$$

4. Scalar Unity &
Scalar Zero

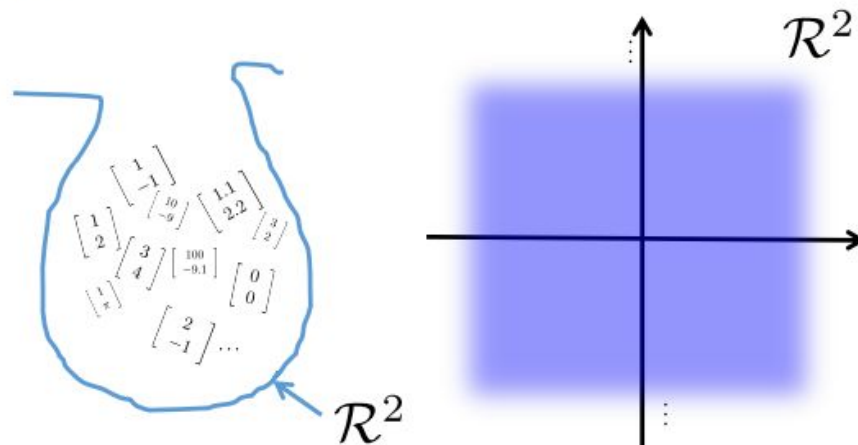
$$1\mathbf{x} = \mathbf{x}$$

$$\mathbf{0}\mathbf{x} = \mathbf{0}, \quad \text{where } \mathbf{0} \text{ is the zero vector of all zeros}$$

Vector Space

$x + y \in V$ with the addition properties

$ax \in V$ with the scalar multiplication properties:



Linear Combination

for vectors $a_1, \dots, a_m$ and scalars $\beta_1, \dots, \beta_m$,

$$\beta_1 a_1 + \dots + \beta_m a_m$$

is a *linear combination* of the vectors

$\beta_1, \dots, \beta_m$ are the *coefficients*

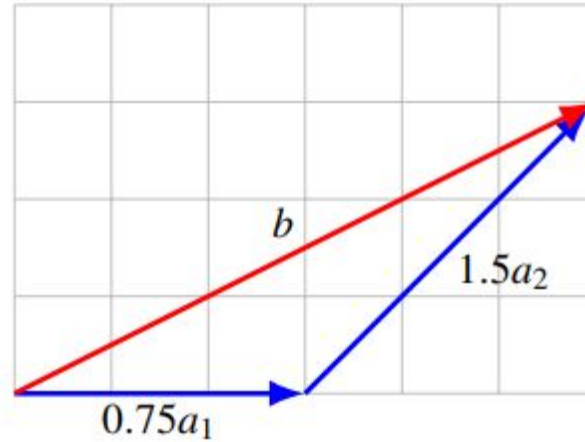
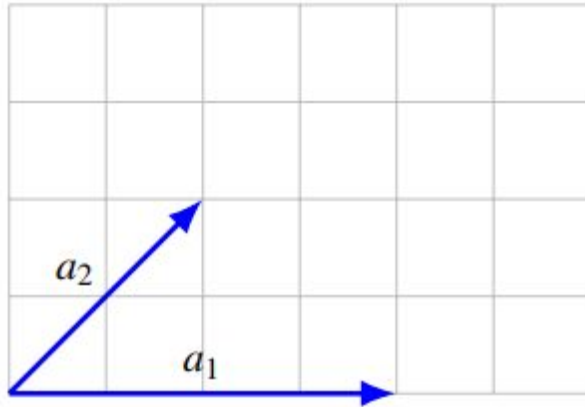
a *very* important concept

a simple identity: for any n -vector b ,

$$b = b_1 e_1 + \dots + b_n e_n$$

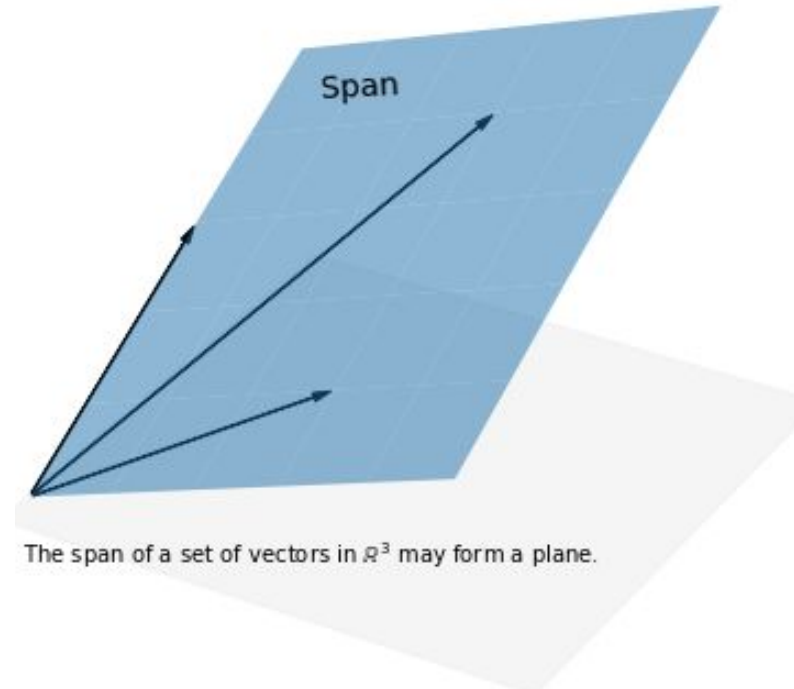
Linear Combination

two vectors a_1 and a_2 , and linear combination $b = 0.75a_1 + 1.5a_2$



Span

- The **span** of vector **u** and **v** in $\mathbb{R}^n$ is the set of all their **linear combinations**. Such that:
$$\text{Span}\{v_1, \dots, v_n\} = \{s_1 v_1 + \dots + s_n v_n \mid s_1, s_2, \dots, s_n \text{ in } \mathbb{R}\}$$
- Span is the subspace of vector space V.

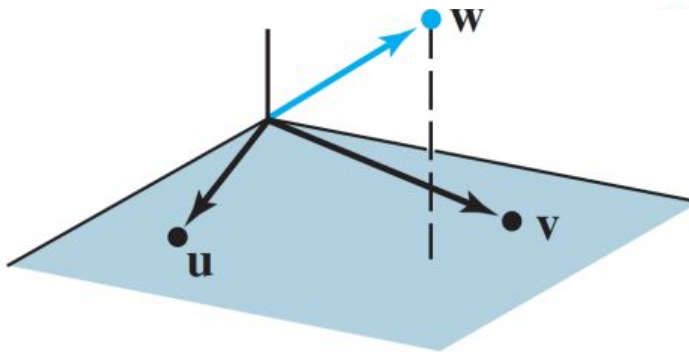


Linear Dependence and Independence

Definition. The vectors $v_1, \dots, v_k$ are **linearly independent** if

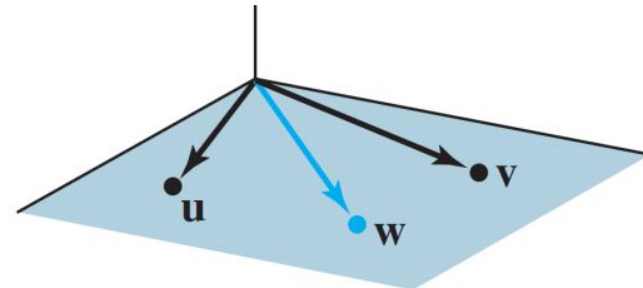
$$\sum_{i=1}^k \alpha_i v_i = 0 \Rightarrow \alpha_i = 0 \text{ for all } i.$$

If a set of vectors is not linearly independent, then they are **linearly dependent**.



Linearly independent

$$w \notin \text{span}(u, v)$$



Linearly dependent
 $w \in \text{span}(u, v)$

Matrix

Review What are Matrices?

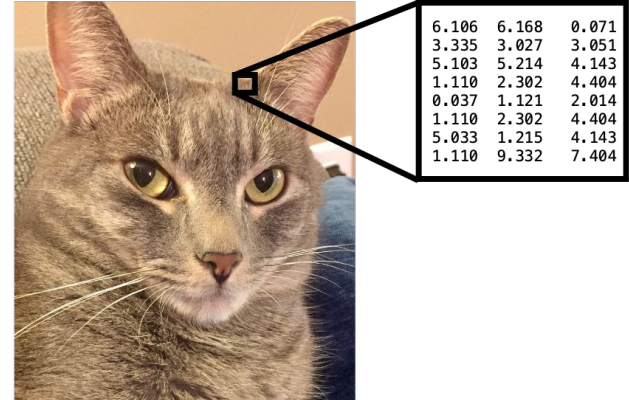
- **Matrix** is a rectangular array of real-valued scalars arranged in m horizontal rows and n vertical columns
 - Each element a_{ij} belongs to the i^{th} row and j^{th} column

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Intuitions on Matrix -1

As a table of data

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520



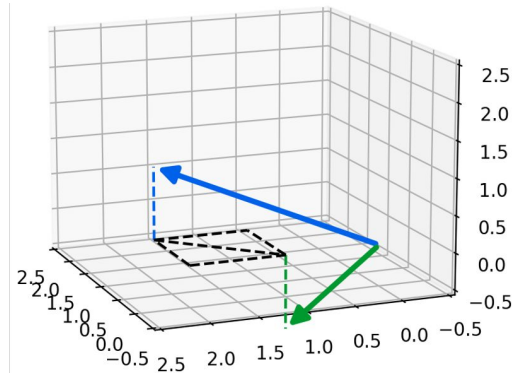
Intuitions on Matrix -2

Collection of vectors (vector of vectors)

- For example, for the following matrix:

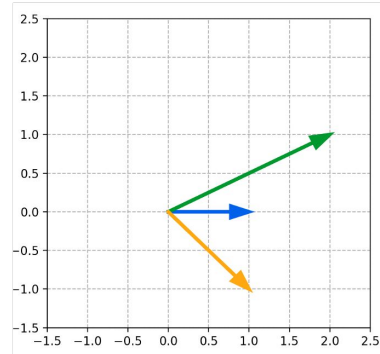
$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

- $A := \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ can be thought of as
 - a **two** **three-dimensional** **row** **vectors** i.e.
 - $\mathbf{a}_{1*} := [1 \ 2 \ 1]$ and $\mathbf{a}_{2*} := [0 \ 1 \ -1]$



- Or as a **three** **two-dimensional** **column** **vectors**:
- $\mathbf{a}_{*1} := \begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $\mathbf{a}_{*2} := \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{a}_{*3} := \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$



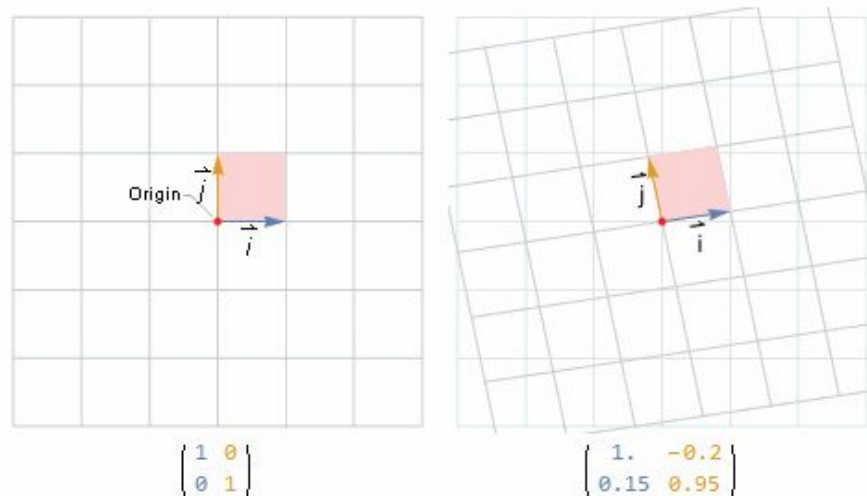
Intuitions on Matrix -3

Interpretation 2: Linear Transformation
Multiplying a $m \times n$ matrix into a $n \times 1$

We can think of this as a linear function in m -dimensional space.

$$Ax \rightarrow y$$

矩阵 $A = \begin{pmatrix} 1 & -2 \\ 1.5 & 0.5 \end{pmatrix}$ 所对应的变换



WeChat ID: meetmath

Image source :
<https://mathematica.stackexchange.com/questions/151238/how-to-use-manipulate-to-show-linear-transformation-of-matrix>

Intuitions on Matrix -4

Intuition 3: As Linear Equations

$$y = Ax$$

System of Linear Equation

$$2.0x + 4.0y + 6.0z = 18$$

$$4.0x + 5.0y + 6.0z = 24$$

$$3.0x + 1y - 2.0z = 4$$

Matrix representation

$$A = \begin{bmatrix} 2.0 & 4.0 & 6.0 \\ 4.0 & 5.0 & 6.0 \\ 3.0 & 1.0 & -2.0 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad b = \begin{bmatrix} 18.0 \\ 24.0 \\ 4.0 \end{bmatrix}$$

Special Matrices.

- Rectangular Matrix:

- Matrices are said to be rectangular when the number of rows is $\neq$ to the number of columns, i.e. $A^{m \times n}$ with $m \neq n$. For instance:

$$A_{2 \times 3} := \begin{bmatrix} 1 & 2 & 3 \\ 5 & 5 & 4 \end{bmatrix}$$

- Square Matrix:

- Matrices are said to be square when the number of rows = the number of columns, i.e. $A^{m \times n}$. For instance:

$$A_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

- Diagonal Matrix:

- Square matrices are said to be diagonal when each of its non-diagonal elements is zero, i.e. for
 - $D = (d_{ij})$, we have $\forall i, j \in n \ i \neq j \Rightarrow d_{ij} = 0$.
- For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

- Upper triangular matrix:

- Square matrices are said to be upper triangular when the elements below the main diagonal are zero i.e. For $D = (d_{ij})$, we have $d_{ij} = 0$, for $i > j$. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 8 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 5 \end{bmatrix}$$

- Lower triangular matrix:

- Square matrices are said to be lower triangular when the elements above the main diagonal are zero i.e. $D = (d_{ij})$, we have $d_{ij} = 0$, for $i < j$. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 0 & 0 \\ 8 & 1 & 0 \\ 4 & 2 & 5 \end{bmatrix}$$

- Identity Matrix:

- A diagonal matrix is said to be the identity when the elements along its main diagonal are equal to one. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Special Matrices.

- **Symmetric Matrix:**

- Square matrices are said to be symmetric if its equal to its transpose, i.e. $\mathbf{A} = \mathbf{A}^T$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 6 \\ 3 & 6 & 1 \end{bmatrix}$$

- **Scalar Matrix:**

- Diagonal matrices are said to be scalar when all the elements along its main diagonal are equal, i.e. $\mathbf{D} = \alpha \mathbf{I}$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- **Null or Zero Matrix:**

- Matrices are said to be null or zero matrices when all its elements equal to zero, which is denoted as $\mathbf{0}_{m \times n}$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- **Equal Matrix:**

- Two matrix are said to be equal if
 - $\mathbf{A}(\mathbf{a}_{ij}) = \mathbf{B}(\mathbf{b}_{ij})$.
- For instance:

$$\mathbf{B}_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\mathbf{A}_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

Special Matrices Design Matrix.

- Structured representation of the input data used for modelling (also known as feature matrix).
- It is a matrix where each row corresponds to a data point (an observation) and each column corresponds to a feature (a variable or predictor) of those data points

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
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3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Matrix Operation: Arithmetic.

Matrix – Matrix Sum:

- Operates element-wise

$$(A \pm B)_{ij} = a_{ij} \pm b_{ij}, \quad A, B \in \mathbb{R}^{m \times n}$$

$$A \pm B = \begin{bmatrix} a_{11} \pm b_{11} & \cdots & a_{1n} \pm b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & \cdots & a_{mn} \pm b_{mn} \end{bmatrix}$$

Matrix – Scalar Multiplication:

- Each element is multiplied by the scalar α (alpha)

$$(\alpha A)_{ij} = \alpha \cdot a_{ij}$$

- Example:

$$\alpha = 2, \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \alpha A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Matrix Operation: Arithmetic.

Matrix – vector Multiplication:

Computes the dot product of each row of A with x

$$y = A \cdot x, \quad A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^{n \times 1}, y \in \mathbb{R}^{m \times 1}$$

Given Matrix A and column vector x ; Compute $A \times x$:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 3 & -1 \\ 1 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3} \text{ and } x = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} \in \mathbb{R}^{3 \times 1}$$

$$\begin{aligned} Ax &= 3 \times \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + (-1) \times \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} + 4 \times \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \times 1 + 3 \times 0 + 3 \times 1 \\ -1 \times 0 - 1 \times 3 - 1 \times 2 \\ 4 \times -2 + 4 \times -1 + 4 \times 1 \end{bmatrix} \\ &= \begin{bmatrix} -5 \\ -7 \\ 5 \end{bmatrix} \in \mathbb{R}^{3 \times 1} \end{aligned}$$

Hadamard Product:

- Multiplies the corresponding elements of A and B

$$(A \circ B)_{ij} = a_{ij} \cdot b_{ij}, \quad A, B \in \mathbb{R}^{m \times n}$$

- Example:

$$A \cdot B = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 \times 1 & 2 \times 3 \\ 1 \times 2 & 4 \times 1 \end{bmatrix} = \begin{bmatrix} 0 & 6 \\ 2 & 4 \end{bmatrix}$$

Matrix Multiplication

Matrix multiplication is an operation that combines two matrices to produce a new matrix. It involves multiplying corresponding elements of rows and columns, then summing the products.

$$\mathbf{A} * \mathbf{B} = \mathbf{C}$$

Conditions: The number of columns in the first matrix (A) must equal the number of rows in the second matrix (B).

Properties:

Associativity: $(\mathbf{A} * \mathbf{B}) * \mathbf{C} = \mathbf{A} * (\mathbf{B} * \mathbf{C})$

Distributivity: $\mathbf{A} * (\mathbf{B} + \mathbf{C}) = \mathbf{A} * \mathbf{B} + \mathbf{A} * \mathbf{C}$

Not Commutative: $\mathbf{A} * \mathbf{B} \neq \mathbf{B} * \mathbf{A}$

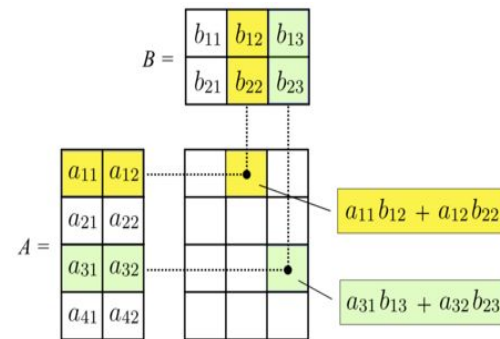
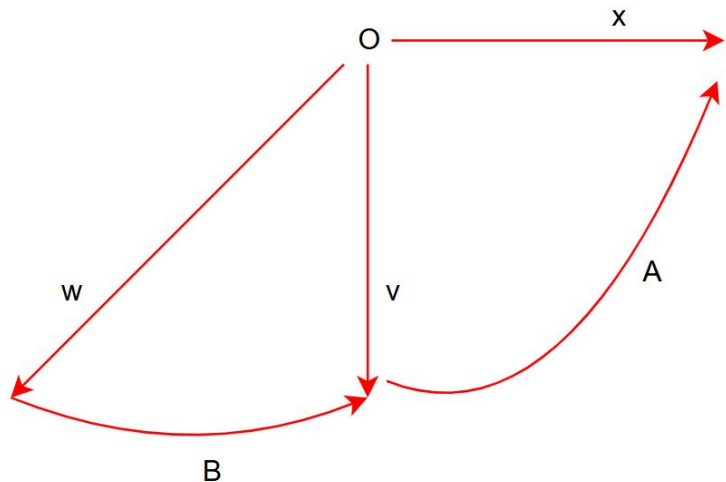


Fig: Schematic representation of Matrix product
Image from Wikipedia

Matrix Multiplication: Geometric Intuition

$$\mathbf{C} = \mathbf{AB}$$

Represents the total resultant transformation obtained by applying **B first** and then **A** to a vector.



Transpose of a Matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 9 \\ 2 & 6 & 10 \\ 3 & 7 & 11 \\ 4 & 8 & 12 \end{bmatrix}^T$$

The transpose of a matrix product has a simple form:

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T.$$

Inverse of a matrix

For a square $n \times n$ matrix $\mathbf{A}$ with rank n , $\mathbf{A}^{-1}$ is its *inverse matrix* if their product is an identity matrix $\mathbf{I}$

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

All invertible matrices are square (dimensional), but not all square matrices are invertible.

Square matrices which are not invertible are called singular.

Inverse of a matrix visualized

$$AA^{-1} = A^{-1}A = I$$

$$A\vec{x} = \vec{v}$$



Pseudo Inverse of Matrices

- For matrices that are not square, the inverse does not exist
 - Therefore, a pseudo-inverse is used
- If $m > n$, then the pseudo-inverse is
 - $\mathbf{A}^\dagger = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ and $\mathbf{A}^\dagger \mathbf{A} = \mathbf{I}$
- If $m < n$, then the pseudo-inverse is
 - $\mathbf{A}^\dagger = \mathbf{A}^T (\mathbf{A} \mathbf{A}^T)^{-1}$ and $\mathbf{A} \mathbf{A}^\dagger = \mathbf{I}$

Determinant of a matrix

How much are **areas** scaled?



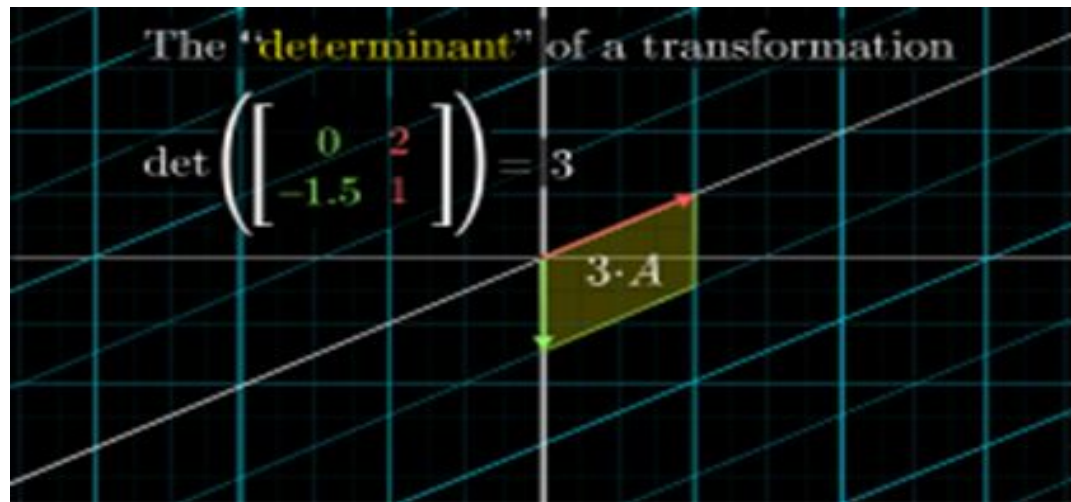
Determinant of the matrix

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

For larger-size matrices the determinant of a matrix is calculated as

$$\det(\mathbf{A}) = \sum_j a_{ij}(-1)^{i+j} \det(\mathbf{A}_{(i,j)})$$

In the above, $\mathbf{A}_{(i,j)}$ is a **minor** of the matrix obtained by removing the row and column associated with the indices i and j

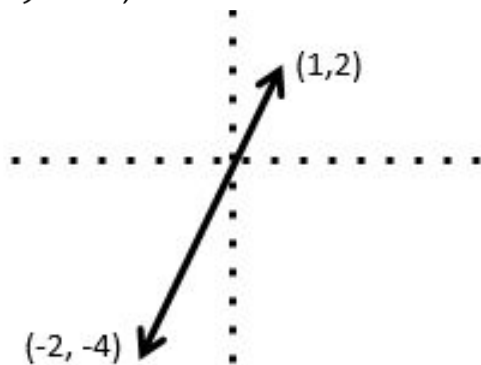


Rank of a matrix

For an $n \times m$ matrix, the *rank* of the matrix is the largest number of linearly independent columns

- The matrix $\mathbf{B}$ from the previous example has $\text{rank}(\mathbf{B}) = 1$, since the two columns are linearly dependent

$$\mathbf{B} = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$$



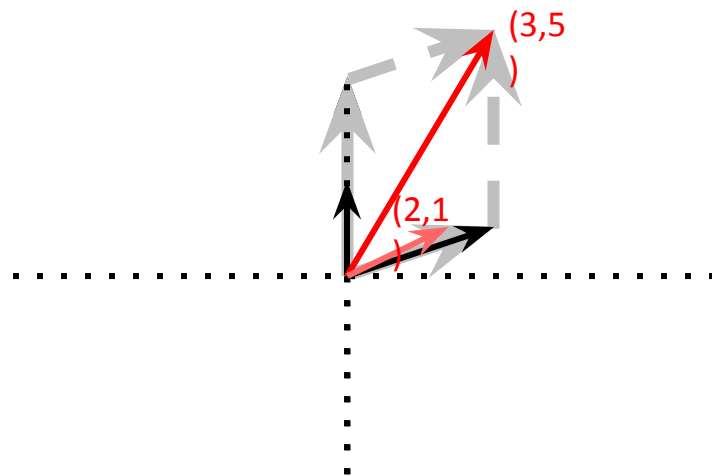
- You can only get vectors along the $(1, 2)$ direction (i.e. outputs live in 1 dimension, so we call the matrix *rank 1*)

Matrix Decomposition-Eigen.

What do matrices do to vectors?

$\overleftrightarrow{M}$

$$\begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$



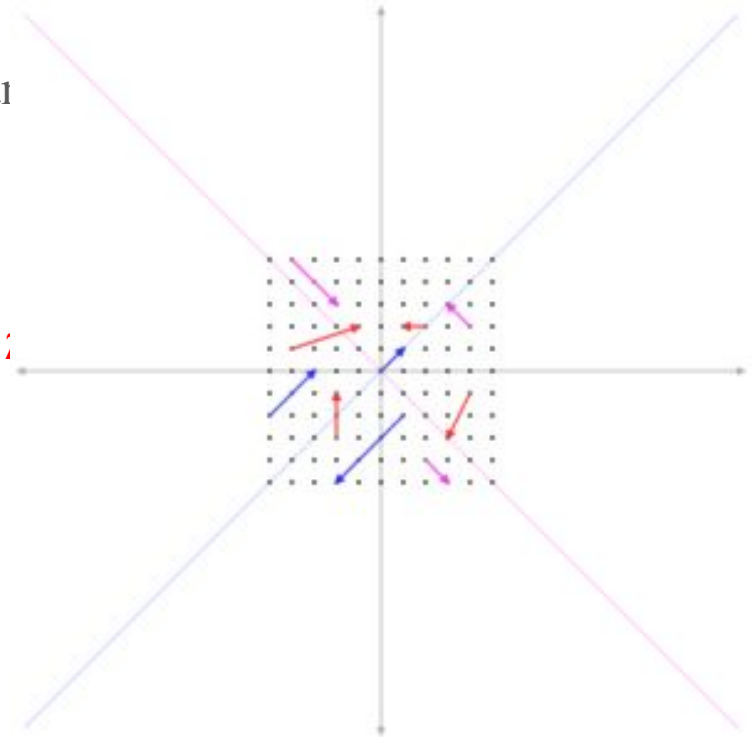
- The new vector is:
 - 1) **rotated**
 - 2) **scaled**

Eigen Vector and Eigen Values.

- Eigen vectors are the vectors i.e. non zero at any linear transformation is applied.
- Only gets scaled up by transformations.

$$\mathbf{Ax} = \lambda \mathbf{x}$$

$\mathbf{x}$ is an eigenvector of $\mathbf{A}$ corresponding to eigenvalue, λ



Computing Eigen Vector and Eigen Values.

- *Eigenvalues* of a square matrix $\mathbf{A}$ are scalars λ and *eigenvectors* are non-zero vectors $\mathbf{v}$ that satisfy

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- Eigenvalues are found by solving the following equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

- The eigenvectors of $\mathbf{A}$ corresponding to λ are the nonzero solutions of $(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = 0$.

Eigen Decomposition.

For a matrix A with n linearly independent eigenvectors: $A = V\Lambda V^{-1}$

V = matrix of eigenvectors

Λ = diagonal matrix of eigenvalues

Key Applications of Eigen Decompositions

- Finding matrix inverse: $A^{-1} = V\Lambda^{-1}V^{-1}$
- Computing determinant: $\det(A)$ = product of eigenvalues
- Analysing matrix properties (positive/negative definite semi-definite, etc.)

Key Concept

Eigenvalues and eigenvectors describe how a matrix stretches or compresses space along specific directions.

Original Space

Eigenvectors v_1, v_2 start as unit vectors (length = 1)

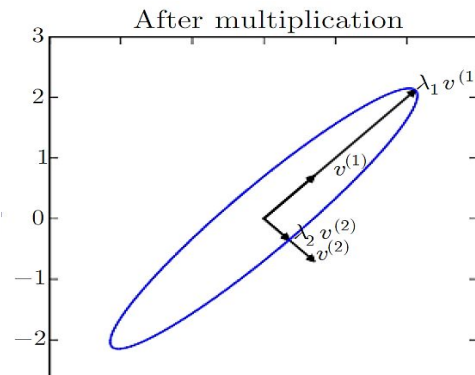
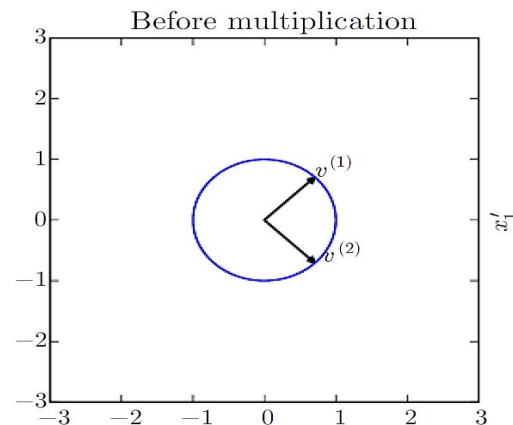
Transformed Space

Vectors are scaled by eigenvalues λ_1, λ_2

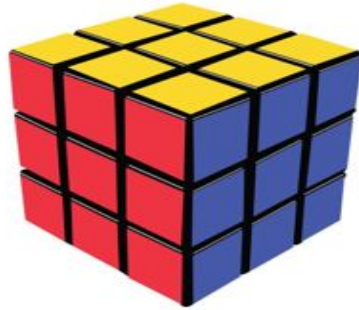
Application Example

In PCA (Principal Component Analysis), we use eigenvectors with the largest eigenvalues to identify the most important directions of variation in data.

pretation.

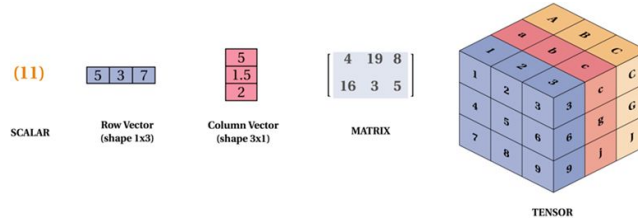


Review : What are Tensors?}



What are Tensors?

- A tensor is a **multidimensional array** and a **generalization** of the concepts of a **vector** and a **matrix**.



- Tensors have many axes, here is a tensor with three axes:

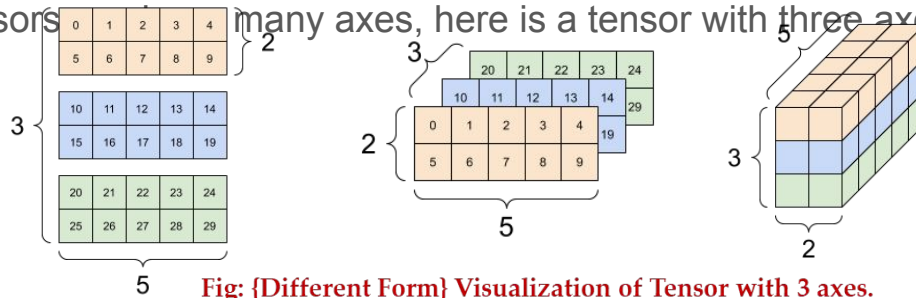
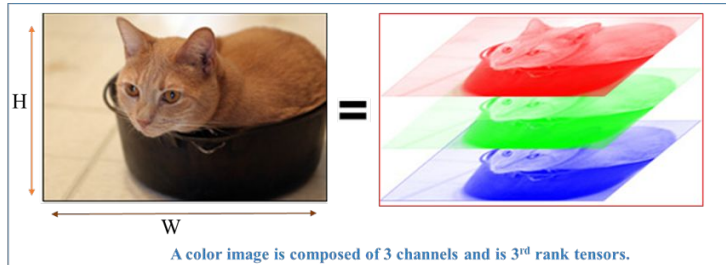


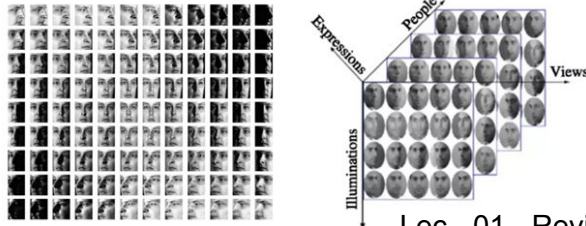
Fig: {Different Form} Visualization of Tensor with 3 axes.

Tensor Example.

- Tensors in DL are Used to represent an image.
 - $\text{image_shape} := \text{Height} \times \text{Width} \times \text{Color Channel (RGB)}$



facial images database is 6th-order tensor

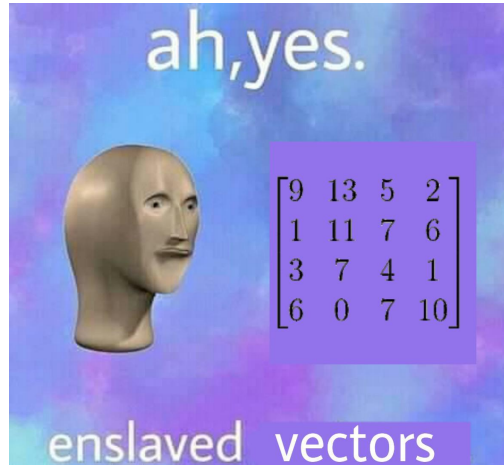


color video is 4th-order tensor



Plan

- Tutorial:
 - Review of Derivative for Deep Learning.
 - Some Hands-on Exercise on Vector, Matrices.
- Workshop – Implement PCA with eigenvalue decomposition for Image Compression Application.



Thank You!!!